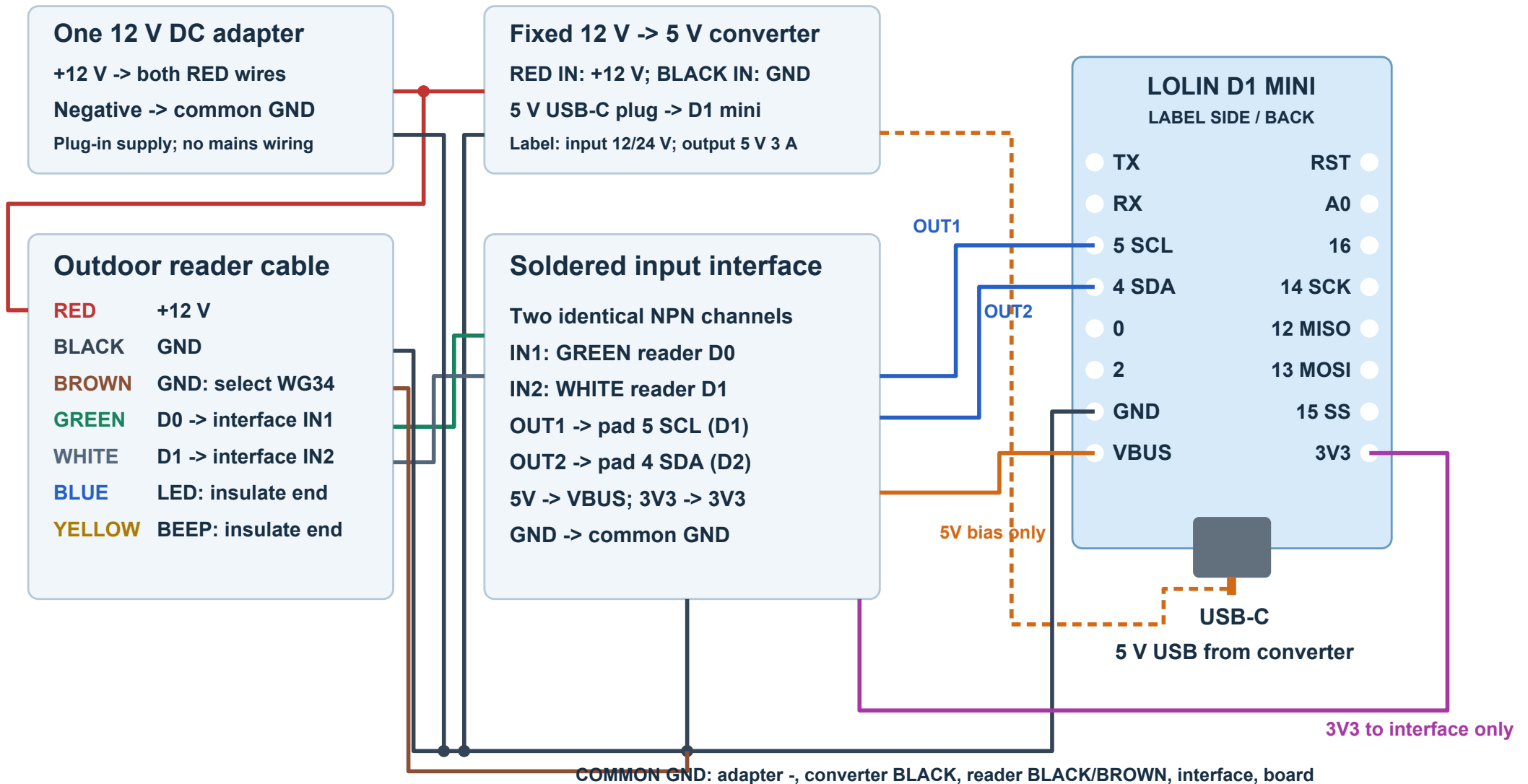


1. Complete wiring and the exact D1 mini pads

ONE wall adapter: 12 V feeds the reader and your pictured 12/24 V-to-5 V, 3 A USB-C converter.



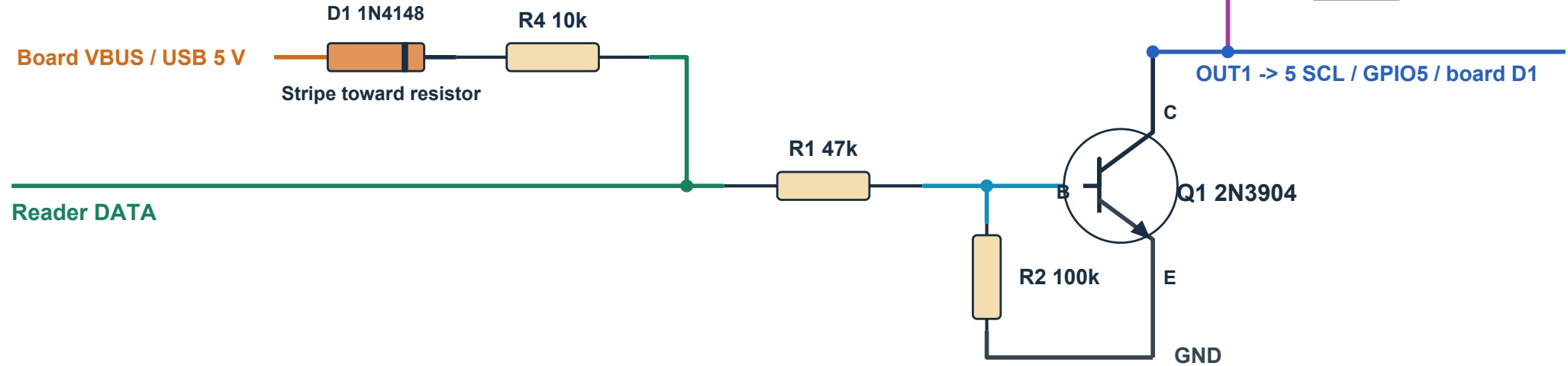
Orient your board by its printed labels. Reader D1 is NOT board D1; follow the mapping above.

12 V never goes to VBUS, 3V3 or GPIO. Leave gate motor/relay wiring to the existing HA gate controller.

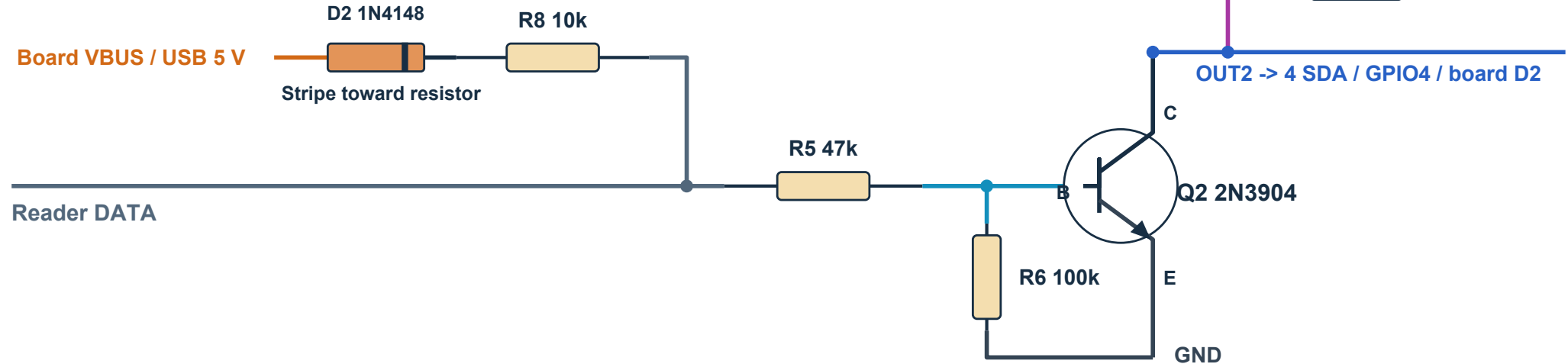
2. Electrical schematic - build both channels

Every filled dot is a soldered junction. Crossing lines without a dot must stay electrically separate.

CHANNEL 1: GREEN / reader D0



CHANNEL 2: WHITE / reader D1

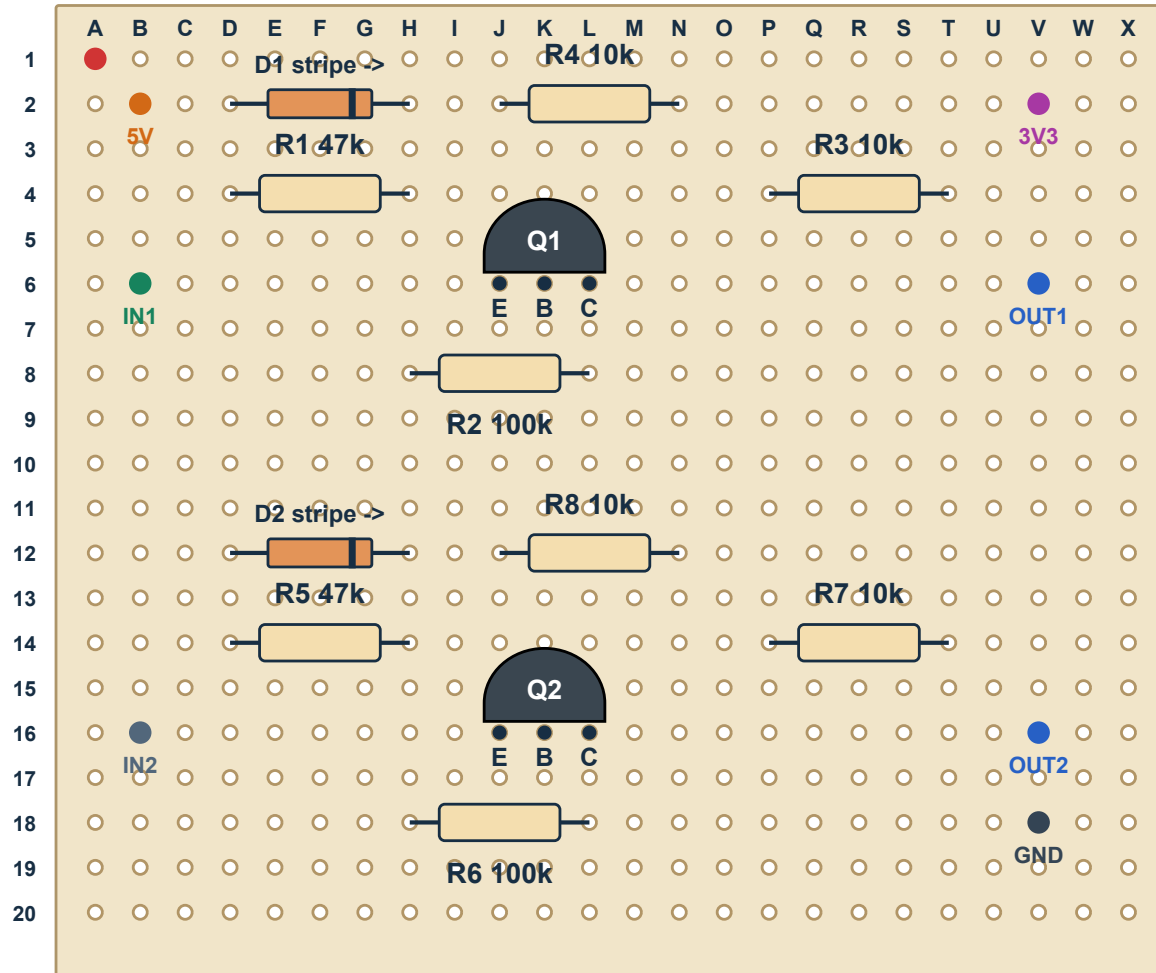


GPIO is connected to the collector junction, not through the collector pull-up resistor.

Both firmware Wiegand pins stay inverted: true. All four resistors in each channel are required here.

3. Component side - placement on perfboard

Use isolated copper pads, 2.54 mm pitch, at least 24 columns x 20 rows. Do NOT use connected stripboard.



Placement / lead endpoints

D1: D2 -> H2 (stripe at H2)

R4 10k: J2 - N2

R1 47k: D4 - H4

Q1: E J6 / B K6 / C L6

R3 10k: P4 - T4

R2 100k: H8 - L8

D2: D12 -> H12 (stripe at H12)

R8 10k: J12 - N12

R5 47k: D14 - H14

Q2: E J16 / B K16 / C L16

R7 10k: P14 - T14

R6 100k: H18 - L18

No pads are connected by default.

Add the wires on page 4 after soldering and trimming the component leads.

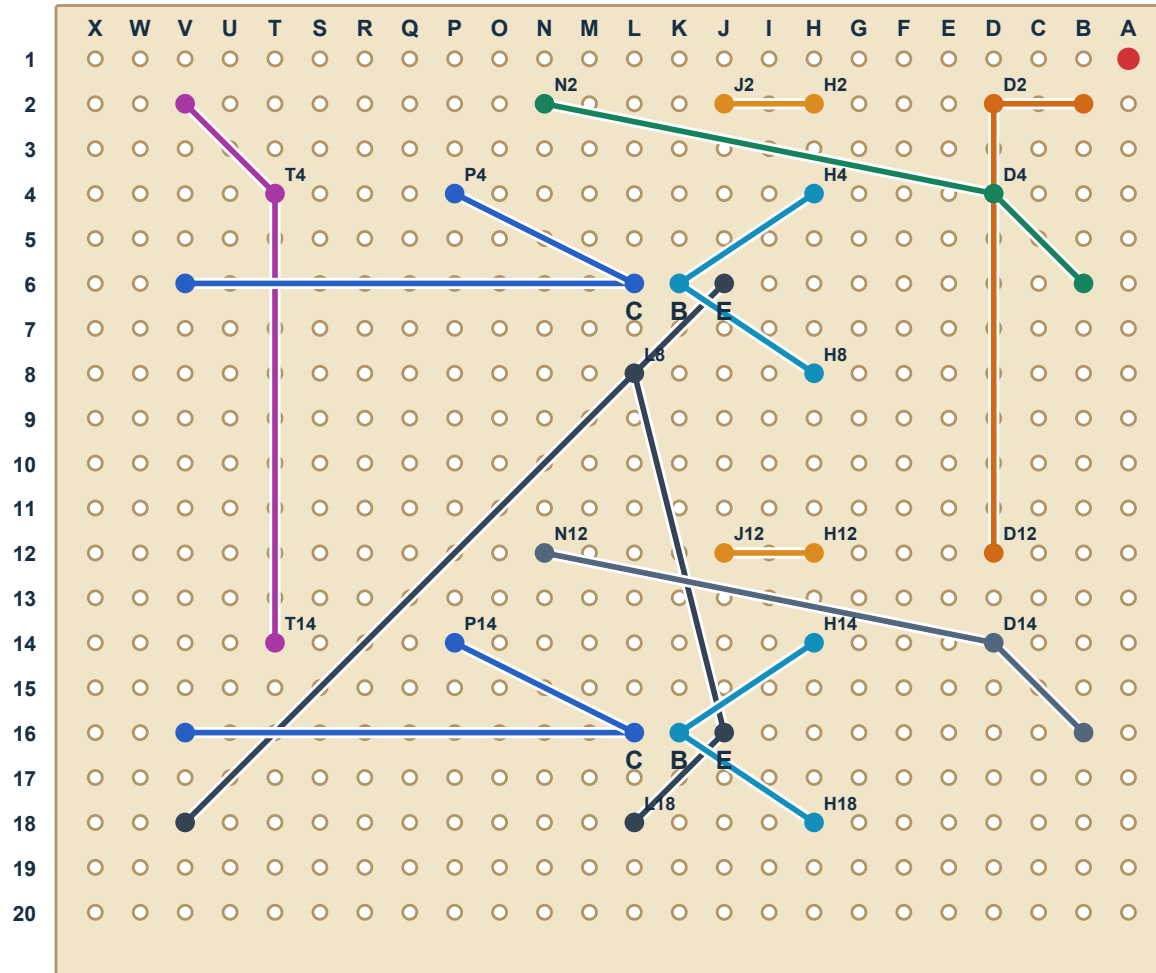
Q1/Q2 must be 2N3904 with the specified E-B-C pin order. Match the supplier's datasheet.

Mark corner A1 in red. Turn the board left-to-right for the solder-side view; keep row 1 at the top.

Q1 flat edge faces row 7; Q2 flat edge faces row 17. Keep all bare component leads separate.

4. Solder side - mirrored wire connections

Use isolated copper pads, 2.54 mm pitch, at least 24 columns x 20 rows. Do NOT use connected stripboard.



Solder together **ONLY** these groups

5V	B2 - D2 - D12
3V3	V2 - T4 - T14
GND	V18 - J6 - L8 - J16 - L18
IN1	B6 - D4 - N2
B1	H4 - K6 - H8
OUT1	V6 - L6 - P4
BIAS1	H2 - J2
IN2	B16 - D14 - N12
B2	H14 - K16 - H18
OUT2	V16 - L16 - P14
BIAS2	H12 - J12

Different groups must never touch.

Wire crossings are NOT junctions.

Use insulation all the way to each pad.

Dots mark solder points. Each color/group forms one electrical net. Route insulated wires as convenient. Strip only 2-3 mm at each wire end. Solder the wire to the existing pad/lead; do not bridge adjacent pads. The numbers and letters identify the SAME holes as page 3, even though this view is mirrored.

5. Parts, soldering order and first power-up

Single-adapter build: use your purchased 12/24 V-input, 5 V / 3 A-output converter with USB-C plug.

Additional parts

- 2 2N3904 NPN, TO-92; matching E-B-C pinout
- 2 1N4148 axial diodes
- 4 10k ohm resistors, 1/4 W, 1% or 5%
- 2 47k ohm resistors, 1/4 W, 1% or 5%
- 2 100k ohm resistors, 1/4 W, 1% or 5%
- 1 Isolated-pad perfboard, $\geq 24 \times 20$ holes (2.54 mm)
- 1 Fixed regulated 12 V DC wall adapter
- 1 Matching DC barrel-to-screw adapter, if needed
- 1 Purchased fixed 5 V / 3 A USB-C converter
- Insulated wire, solder, flux, heat-shrink, enclosure

12 V supply capacity must cover reader + converter.

5 V / 3 A is output capacity, not constant current draw.

Resistor identification (4-band, gold tolerance)

10k: brown - black - orange - gold

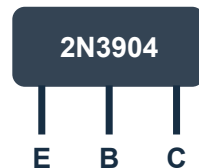
47k: yellow - violet - orange - gold

100k: brown - black - yellow - gold

2N3904 FRONT VIEW

Flat face toward you; legs down.

Supplier pin order must match.



Build with adapter and USB unplugged

1. Solder the D1 mini headers, or use short wires directly on the labelled pads shown on page 1.
2. Insert resistors and diodes per page 3. Resistors have no polarity; diode stripes must face H2/H12.
3. Fit both 2N3904s. Spread leads gently to the holes; keep E, B and C separate. Solder; trim the leads.
4. Add INSULATED wires on the copper side using the page 4 groups. No stripboard / common rows.
5. Connect the seven interface wire pads below.
6. +12 V -> reader RED + converter RED. Negative joins converter BLACK, reader BLACK/BROWN, GND.
7. Insulate BLUE/YELLOW. Keep controller, converter and interface indoors; secure all cables.
8. Inspect for shorts. Check actual converter label says 12/24 V input, 5 V output before power-up.
9. Flash with PC USB; unplug converter USB first. Then swap PC USB for converter USB and power 12 V.
10. Check Last Frame Bits = 34 and Last Card in HA. Try a notification before enabling gate actions.

Interface pads -> external wires

B2 -> board VBUS; V2 -> board 3V3; V18 -> GND

B6 -> GREEN; B16 -> WHITE

V6 -> board 5 SCL; V16 -> board 4 SDA

Without a meter, use fixed regulated supplies, not an adjustable buck converter. Visual inspection is limited.

This input circuit is intended for ordinary 0-12 V Wiegand logic; it is not a certified outdoor surge protector.

Sources: your reader label; WEMOS D1 mini documentation; onsemi 2N3903/2N3904; Vishay 1N4148 datasheets.